Task1

Objective

This challenge aims to assess contestants' ability to integrate and deploy various services within the Google Cloud Platform (GCP). Contestants must complete a full application based on Cloud Run, Cloud SQL, and Cloud Storage while automating deployment through Cloud Build, Cloud Deploy, and Cloud Scheduler.

Note: Please refer to Table 1 for configuration of required resources! If it is a mentioned service, you can name it yourself

Contestants must complete the following tasks:

1. Create Cloud SQL and Set Up Tables

- Create a Cloud SQL instance.
- Use the SQL script provided in `gs://task1-source-code/sql.txt` to create tables and columns.
- Ensure Cloud SQL can be connected to the Cloud Run service.

2. Process Cloud Storage Source Files

- `gs://task1-source-code`: Files in this folder **cannot be modified**, contestants can only read these resources.
- `gs://task1-source-code-modify`: Files in this folder **can be modified**, contestants must adjust the contents as needed.

3. Configure Cloud Build and Cloud Deploy

- Use Cloud Build to read the source code and Dockerfile to build the image from `gs://task1-source-code`.
- Push the built image to Artifact Registry.
- Cloud build needs to be able to create images and create releases to create new Cloud runs

4. Configure Cloud Scheduler

- Create a Cloud Scheduler job to run **every hour**.
- Cloud Scheduler must trigger Cloud Build to rebuild and redeploy Cloud Run.

5. Configure Cloud Run and Connect to VPC and Cloud SQL

- Configure Cloud Deploy to deploy the latest image to Cloud Run and set the latest version tag.
- Deploy the Cloud Run service and ensure it can connect to Cloud SQL.
- Cloud Run must use a VPC internal IP to access Cloud SQL (not a public IP).
- Ensure Cloud Run sets database connection information via environment variables.
- Set up Cloud Deploy to support rollback, allowing reversion to the previous version.
- You can refer to `gs://task1-source-code-modify/yaml-sample.yaml` to help you complete the VPC and environment variable settings.

6. Set Up Cloud Load Balancer and Cloud Storage Website

- Host index.html as the website entry point using Cloud Storage.
- Provide public access via Cloud Load Balancer and Cloud CDN.
- Ensure the website forwards requests to Cloud Run and Cloud Run stores request data in Cloud SQL.

Completion Criteria

Contestants must successfully complete the following tasks and pass the tests:

1. **Cloud SQL functions correctly**: Able to access SQL and read/write data.
2. **Cloud Run operates normally**: The service can respond to API requests and interact with Cloud SQL.

3. **Cloud Build & Cloud Deploy function correctly**: Cloud Build correctly builds the image, and Cloud Deploy automatically deploys it.
4. **Cloud Scheduler functions correctly**: It triggers Cloud Build every hour to redeploy Cloud Run.
5. **Cloud Load Balancer and Cloud Storage function correctly**: The website displays index.html and interacts with Cloud Run through the frontend.

Contestants must complete all configurations as required and submit the final solution within the given timeframe.

Table 1

Network	
VPC name	task1-vpc
Subnet name	task1-sb
Region	us-central1
IPv4 range	10.0.0.0/16

Cloud SQL	
Database Engine	MySQL
Cloud SQL edition	Enterprise
Preset	Sandbox
Instance ID	task1-sql
Username	root
User Password	password
region	us-central1
Machine configuration	2 vCPU, 8 GB
Storage type	SDD,10GB
Network	task1-vpc

Load Balancer	
Name	task1-ealb
Frontend Protocol	HTTP
Frontend Port	80

Backend Service Name	task1-bked